

Topics: Review of derivatives and integrals.

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1. Find the derivative of $f(x) = \cos(x^2 + e^x)$.
2. Given that $x \ln(y) + e^y = 3$, find $\frac{dy}{dx}$ by implicit differentiation.
3. Suppose that $f(x)$ is a differentiable function such that $f(1) = 7$ and $f'(1) = 4$. Find an equation of the tangent line to $y = \sqrt{4 + 3f(x)}$ at $x = 1$.

4. Find all critical points of

$$f(x) = x^3 - 3x^2 - 9x + 2$$

and determine whether each is a local maximum or a local minimum.

5. Evaluate the indefinite integrals.

(a) $\int e^{\tan x} \sec^2(x) dx$

(b) $\int (2x^4 + 1) \ln(x) dx$

6. Evaluate the definite integrals.

(a) $\int_0^{\pi/2} \sin^2 \theta \cos \theta d\theta$

(b) $\int_0^3 \frac{x}{\sqrt{x^2 + 1}} dx$

7. A particle is moving with acceleration at time t given by

$$a(t) = 3 \cos t - 2 \sin t.$$

Given that $s(0) = 0$ and $v(0) = 4$, determine the position of the particle $s(t)$ at any time t .

8. Consider the region R bounded by the curves $x = y^2$ and $y = \frac{1}{2}x$.

(a) Sketch the region R .

(b) Find the area of the region R .

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Differentiation Formulas

$$\frac{d}{dx}(c) = 0$$

$$\frac{d}{dx}(b^x) = b^x \ln b$$

$$\frac{d}{dx}(f(x) + g(x)) = f'(x) + g'(x)$$

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(cf(x)) = cf'(x)$$

$$\frac{d}{dx}(f(x) - g(x)) = f'(x) - g'(x)$$

$$\frac{d}{dx}(e^x) = e^x$$

The Product Rule

$$\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$$

The Quotient Rule

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$$

The Chain Rule

$$\frac{d}{dx}(f(g(x))) = f'(g(x))g'(x)$$

Derivatives of Trigonometric Functions

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

The Chain Rule for the power of a function

$$\frac{d}{dx}[f(x)^n] = n[f(x)]^{n-1} f'(x)$$

Derivatives of Inverse Trigonometric Functions

$$\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx}(\arccos x) = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\operatorname{arccot} x) = -\frac{1}{1+x^2}$$

Derivatives of Logarithmic Functions

$$\frac{d}{dx}(\log_b x) = \frac{1}{x \ln b}$$

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

$$\frac{d}{dx}[\ln f(x)] = \frac{f'(x)}{f(x)}$$

Integration Formulas

a. If $n \neq -1$, $\int x^n dx = \frac{x^{n+1}}{n+1} + C$

b. $\int e^x dx = e^x + C$

c. $\int \sin x dx = -\cos x + C$

d. $\int \sec^2 x dx = \tan x + C$

e. $\int \sec x \tan x dx = \sec x + C$

f. $\int \sec x dx = \ln |\sec x + \tan x| + C$

g. $\int \tan x dx = \ln |\sec x| + C$

h. $\int \frac{1}{x} dx = \ln |x| + C$

i. $\int b^x dx = \frac{b^x}{\ln b} + C$

j. $\int \cos x dx = \sin x + C$

k. $\int \csc^2 x dx = -\cot x + C$

l. $\int \csc x \cot x dx = -\csc x + C$

m. $\int \csc x dx = -\ln |\csc x + \cot x| + C$

n. $\int \cot x dx = \ln |\sin x| + C$

where C is any real constant.

The Substitution Rule: This rule is useful when the integrand contains a function and (a constant multiple of) its derivative. Let $u = g(x)$, so $du = g'(x) dx$. Then

$$\int f(g(x)) g'(x) dx = \int f(u) du.$$

Integration by Parts: This rule is useful for integrals that look like a product of two functions. Choose u and dv from the integrand so that $du = u' dx$ and $v = \int dv$. Then

$$\int u dv = uv - \int v du.$$